DeNovo: A Neuro-Symbolic Framework for Verifiable LLM Explanations in Molecular Property Prediction

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***Abstract*—LLMs are increasingly used to translate GNN internal representations into natural language explanations, but these explanations are frequently *unfaithful*—they cite chemical features that did not drive the model's prediction. Existing XAI methods produce attribution scores but provide no mechanism to verify that an LLM's narrative reflects those attributions. We address this gap with DeNovo, a neuro-symbolic framework that enforces causal faithfulness between GNN attention and LLM-generated explanations, evaluated on molecular toxicity prediction. DeNovo's core contribution is the Faithful XAI Engine: a "Reject-and-Retry" validation loop that (1) extracts attention-grounded chemical evidence via a SMARTS-based Substructure Mapper, (2) constrains LLM generation to that evidence, and (3) validates each explanation through counterfactual intervention before presenting it to the user. Faithfulness is quantified as the geometric mean of a *grounding* score (cited features carry high model attention) and a *causal-consistency* score (removing cited features changes the prediction). On the Tox21 benchmark, our Attention-GIN achieves a test ROC-AUC of 0.802 (validation 0.837), and the reject-and-retry mechanism raises mean faithfulness from 0.020 (unconstrained LLM) to 0.865 over the full test population (bootstrap 95% CI $[0.815, 0.910]$), rejecting 13.5% of explanations as unfaithful. Causal faithfulness validation is the primary scientific contribution; we demonstrate the framework on molecular toxicity prediction using the Tox21 benchmark; the surrounding production platform demonstrates practical applicability.**

***Index Terms*—drug discovery, ADMET prediction, explainable AI, graph neural networks, large language models, faithfulness validation, molecular toxicity, counterfactual analysis**

I. Introduction

The pharmaceutical industry is facing a productivity crisis known as "Eroom's Law"—the empirical observation that drug discovery is becoming exponentially slower and more expensive over time, despite significant technological improvements [1]. The cost to bring a new drug to market now exceeds $2.5 billion USD, with the entire process taking 10-15 years from initial discovery to regulatory approval. The single largest contributor to this attrition involves safety failures: nearly 30% of drug candidates fail in Phase I clinical trials due to unmanageable toxicity, and another 20-30% fail due to poor pharmacokinetics (Absorption, Distribution, Metabolism, Excretion, and Toxicity—collectively known as ADMET).

Historically, ADMET profiling was performed using *in vivo* animal models or *in vitro* cellular assays. While accurate, these methods are resource-intensive, low-throughput, and increasingly face ethical scrutiny. In recent years, computational approaches have emerged as promising alternatives for early-stage screening. Graph Neural Networks (GNNs) [2], [3], which naturally respect the non-Euclidean topology of molecular graphs, have achieved state-of-the-art accuracy in predicting molecular properties from structure alone. Studies on the MoleculeNet benchmark [4] have demonstrated that Message Passing Neural Networks (MPNNs) consistently outperform traditional fingerprint-based methods like Extended Connectivity Fingerprints (ECFP) with Random Forests or Support Vector Machines.

However, despite their accuracy, the adoption of GNN-based models in real-world medicinal chemistry pipelines remains limited. The primary barrier is the *Interpretability Gap*. A medicinal chemist cannot act on a scalar prediction (e.g., "$P(Toxic) = 0.95$") without understanding the structural basis of that prediction. Does the model flag the molecule because of a known toxicophore (e.g., a nitro group susceptible to nitroreduction), or is it reacting to a benign scaffold that happens to be overrepresented in the toxic class of the training data? In safety-critical decision-making, being "right for the wrong reasons" is unacceptable.

Explainable AI (XAI) methods for graphs, such as GNNExplainer [5], PGExplainer [6], and SubgraphX [7], attempt to bridge this gap by highlighting important subgraphs or edges. While mathematically rigorous, these explanations are often presented as heatmaps that lack semantic clarity. A heatmap highlighting atoms 3, 5, and 7 does not communicate to a chemist why these atoms are important—e.g., "potential for metabolic activation via epoxidation."

The advent of Large Language Models (LLMs) like GPT-4 [8] and LLaMA [9] offers a potential solution: generating fluent natural language justifications that are directly interpretable by domain experts. However, integrating LLMs into scientific workflows introduces a new and dangerous risk: *hallucination*. LLMs are probabilistic token predictors

trained on internet-scale corpora, not causal reasoning engines grounded in physical chemistry. When asked to explain a toxicity prediction, an LLM might generate a chemically plausible but factually incorrect narrative—citing functional groups that are not present in the molecule or toxicity mechanisms that are irrelevant to the model's actual decision boundary.

A. The Trust Crisis in AI-Driven Drug Discovery

Recent studies have quantified the "hallucination problem" in scientific LLMs. Galactica, a model trained specifically on scientific literature, was retracted within days of release due to generating plausible-sounding but incorrect statements [10]. ChemLLM and similar chemistry-focused models exhibit similar issues, inventing reaction mechanisms or citing nonexistent literature. In the context of toxicity prediction, a hallucinated explanation could lead a chemist to deprioritize a promising drug candidate based on false safety concerns, or worse, advance a dangerous compound based on false reassurance.

This creates a fundamental tension:

- **GNNs**: High accuracy, but opaque ("black box").
- **XAI methods**: Mathematically grounded, but semantically unclear.
- **LLMs**: Semantically clear, but potentially hallucinated.

B. Our Contribution: Faithful Neuro-Symbolic Verification

To address these challenges, we present **DeNovo**, a comprehensive, multi-task AI platform designed for trusted early-stage drug discovery. Unlike disjoint research scripts, DeNovo integrates predictive modeling for six critical ADMET endpoints into a single unified system accessible via a modern web interface.

Our core scientific contribution is the **Faithful XAI Engine**, a neuro-symbolic architecture that enforces mathematically verifiable alignment between the GNN's internal representations and the LLM's textual output. **Critically, we do not treat attention weights as causal explanations by default.** Instead, attention serves as a *candidate evidence generator*—a proposal mechanism whose validity must be verified through counterfactual intervention. We introduce a **Counterfactual Validation** mechanism that tests whether removing the cited toxicophore actually changes the prediction. Only claims that pass this causal test ($\Delta y > \delta$) are accepted, providing a causal, not just correlational, guarantee.

Our contributions are:

1) **Faithful XAI Engine (Core Contribution)**: A neuro-symbolic validation framework that ensures LLM-generated explanations are causally grounded in the GNN's decision process, not hallucinated post-hoc rationalizations.
2) **Attention-GIN Architecture**: A novel GNN variant with 5 GINEConv layers, 300-dimensional embeddings, and a learnable Global Attention Pooling mechanism that produces interpretable per-atom importance scores.
3) **Counterfactual Validation**: A mechanism that tests whether removing cited toxicophores actually changes predictions, providing causal rather than correlational guarantees.
4) **Supporting Infrastructure**: Substructure Mapper (45 SMARTS patterns) with a size-invariant adaptive attention cutoff, an ADMET Rule Engine (decision-support flagging), and a production web platform demonstrating real-world applicability.

The remainder of this paper demonstrates that counterfactual validation substantially improves explanation faithfulness while maintaining competitive predictive performance.

II. Related Work

A. Graph Neural Networks for Molecular Property Prediction

The representation of molecules as graphs—where atoms are nodes and chemical bonds are edges—is natural for deep learning. Early approaches used fixed molecular descriptors such as Extended Connectivity Fingerprints (ECFP) [11] fed into fully connected networks. While effective for some tasks, these methods discard spatial information and cannot learn task-specific representations.

The introduction of Message Passing Neural Networks (MPNNs) [3] enabled end-to-end learning of molecular representations. In this paradigm, information flows between neighboring atoms through iterative message-passing steps, allowing the model to capture complex structural patterns. Several variants have been proposed:

- **Graph Convolutional Networks (GCN)** [12]: Apply spectral convolutions in the graph domain.
- **Graph Attention Networks (GAT)** [13]: Use attention mechanisms to weight neighbor contributions.
- **Graph Isomorphism Networks (GIN)** [2]: Achieve maximal expressive power under the Weisfeiler-Lehman test.
- **GINE (GIN with Edge features)**: Extend GIN to incorporate bond type and stereochemistry.

DeepTox [14], the winning solution of the Tox21 Data Challenge, demonstrated that deep multitask learning could outperform traditional methods. ChemProp [15] introduced directed message passing and achieved state-of-the-art results on MoleculeNet. More recently, Transformer-based architectures like Graphormer [16] have shown promise, though at significantly higher computational cost.

Despite these advances, a critical limitation remains: most architectures use global pooling operations (sum, mean, or max) to aggregate node features into a graph-level representation. This pooling operation destroys the node-level granularity required for identifying specific toxicophores. Our Attention-GIN addresses this by replacing global pooling with a learnable attention mechanism that preserves per-atom importance information.

B. Explainability in Graph Learning

Post-hoc explainability methods for GNNs can be categorized into:

Perturbation-based methods: GNNExplainer [5] learns a soft mask over edges that maximizes mutual information

between the original prediction and the masked subgraph. PGExplainer [6] trains a separate network to generate these masks. SubgraphX [7] uses Monte Carlo Tree Search to find connected subgraphs with high Shapley values.

Gradient-based methods: Integrated Gradients [17] and Saliency Maps compute input attributions based on gradients of the output with respect to input features.

Attention-based methods: GAT and similar architectures produce attention weights as a byproduct of inference. However, attention weights do not always correlate with importance [18].

Despite their utility, these methods share two fundamental limitations:

1) **Semantic Ambiguity**: They return subsets of atoms or edges without explaining the underlying mechanism. A chemist seeing "atoms 3, 5, 7 are important" gains no insight into *why*.
2) **Instability**: Small perturbations to the input molecule can produce radically different explanations, undermining trust.

Our approach addresses the first limitation by using LLMs to translate structural importance into semantic chemical reasoning. We address the second through rigorous causal validation.

C. Large Language Models in Chemistry

LLMs have demonstrated remarkable few-shot capabilities across scientific domains [19]. Chemistry-focused models include:

- **Galactica** [10]: Trained on 106 million scientific documents, but retracted due to hallucination issues.
- **ChemBERT** [20]: BERT pretrained on SMILES strings.
- **MolT5** [21]: Sequence-to-sequence model for molecule captioning.
- **LLaMA-3** [9]: General-purpose LLM with strong instruction-following capabilities.

The application of LLMs to *explain* discriminative GNN predictions is nascent. Most existing approaches operate in a "post-hoc" manner: the LLM is shown the molecule and prediction and asked to rationalize it. This decouples the explanation from the model's actual internal mechanics, creating opportunities for hallucination.

Our framework differs fundamentally by *constraining* LLM generation based on the GNN's internal attention states. The LLM receives not just the molecule and prediction, but a structured "evidence list" of substructures with high attention weights. Furthermore, every explanation is validated through counterfactual testing before being shown to the user.

D. Faithfulness of LLM Explanations

A growing body of work shows that natural-language explanations produced by LLMs are frequently *unfaithful*—they do not reflect the factors that actually drove the model's output. Turpin et al. [22] demonstrate that chain-of-thought rationales can systematically misrepresent the true cause of a prediction, and Madsen et al. [23] find that LLM self-explanations often fail counterfactual faithfulness tests. In parallel, the retrieval-augmented generation literature has begun to separate *correctness* from *faithfulness*, observing that an answer can be correct while its cited justification is not the one the model used [24]. Our work operationalizes this distinction in a neuro-symbolic setting: rather than trusting the LLM's narrative, we treat it as an untrusted translator whose every claim must survive a causal intervention on the underlying GNN. Methodologically, our grounding score is analogous to attribution-overlap metrics and our causal-consistency score to deletion-based faithfulness tests from the ERASER benchmark [25]; unlike prior work, we require both to hold and enforce them through counterfactual generation. Counterfactual explanation for GNNs has been studied structurally [26], but to our knowledge has not been used to *validate* LLM-generated chemical narratives.

III. System Architecture

DeNovo is designed as a modular prototype implementation consisting of three primary layers (Fig. 1).

A. Frontend Layer

The user interface is built with React.js and features:

- **Molecule Input**: SMILES string entry, chemical name lookup (via PubChem API), and text description processing (via AI).
- **Visualization**: Real-time 2D molecular structure rendering using RDKit.js with attention heatmap overlay.
- **ADMET Dashboard**: Unified view of all six prediction endpoints with confidence scores and risk indicators.
- **Explanation Panel**: Natural language explanation with highlighted toxicophores and faithfulness score.
- **Batch Processing**: CSV upload for screening molecular libraries (up to 1000 compounds).
- **AI Chat**: Conversational interface powered by Groq LLaMA-3 for querying toxicity mechanisms.

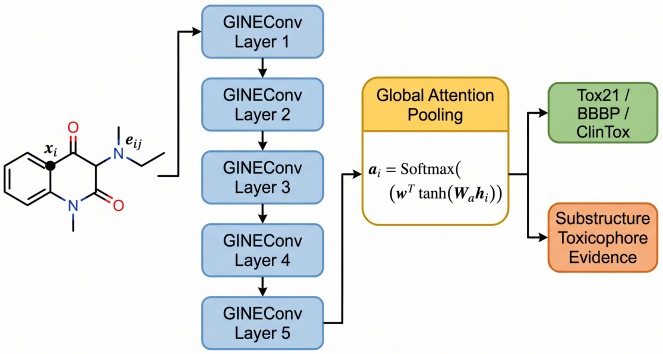


Fig. 1. The DeNovo Platform Interface. The dashboard unifies predictive modeling, 3D/2D visualization, and automated explainability with safety guarantees. Red halos on the molecule indicate attention-based evidence verified by the Faithfulness Validator.

B. Backend Layer

The backend exposes REST APIs for model inference, molecular preprocessing, and explanation generation.

- **REST API**: Endpoints for single prediction, batch prediction, chemical name resolution, and AI chat.
- **Caching**: LRU cache for predictions (configurable size, default 1000 entries) to reduce latency.
- **Rate Limiting**: Protection against API abuse (100 requests/minute per IP).

C. XAI Controller

The explainability engine orchestrates the interaction between the GNN predictor and the LLM explainer:

1) **Substructure Mapper**: Converts attention weights to SMARTS-matched toxicophores.
2) **Evidence Extractor**: Selects top-k substructures with attention $> \tau$ as the "evidence list."
3) **Constrained Explainer**: Prompts LLM with evidence list and strict instructions to cite only verified features.
4) **Counterfactual Generator**: Creates chemically valid perturbations for causal testing.
5) **Faithfulness Validator**: Computes grounding and causal consistency scores; rejects unfaithful explanations.

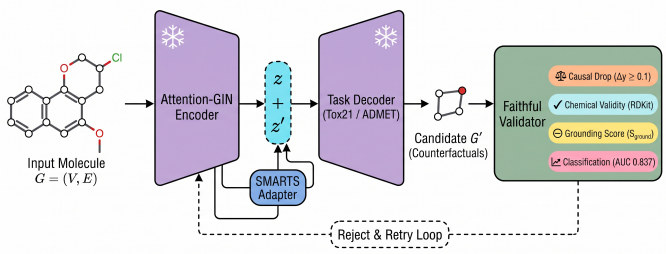


Fig. 2. The Attention-GIN Architecture. Five layers of GINEConv (message passing) with 300-dimensional embeddings process the molecular graph. The Global Attention Pooling mechanism learns to assign relative importance to atoms, producing candidate evidence for the explainability engine.

IV. Methodology

A. Preliminaries

Let a molecule be represented as a graph $G = (V, E)$, where $V = \{v_1, \ldots, v_N\}$ is the set of N atoms and $E \subseteq V \times V$ is the set of chemical bonds. Each atom v_i is associated with a feature vector $\mathbf{x}_i \in \mathbb{R}^{119}$ encoding:

- Atomic number (one-hot, 119 elements including mask tokens)
- Chirality tag (R/S/unspecified)

Each bond $(v_i, v_j) \in E$ has a feature vector $\mathbf{e}_{ij} \in \mathbb{R}^5$ encoding:

- Bond type (single, double, triple, aromatic, self-loop)
- Bond direction (none, endupright, enddownright)

B. Attention-GIN Encoder

We utilize a Graph Isomorphism Network with Edge features (GINE) as the backbone. Standard GINs achieve maximal expressive power under the Weisfeiler-Lehman graph isomorphism test [2], meaning they can distinguish any pair of non-isomorphic graphs.

The node embedding for atom v at layer k is updated as:

$$\mathbf{h}_v^{(k)} = \text{MLP}^{(k)}\Big((1+\epsilon^{(k)}) \cdot \mathbf{h}_v^{(k-1)} + \sum_{u \in \mathcal{N}(v)} \phi(\mathbf{h}_u^{(k-1)}, \mathbf{e}_{uv})\Big) \tag{1}$$

where $\epsilon^{(k)}$ is a learnable scalar, $\mathcal{N}(v)$ denotes the neighbors of v, and ϕ is a message function that combines node and edge features:

$$\phi(\mathbf{h}_u, \mathbf{e}_{uv}) = \mathbf{h}_u + \text{Embed}_{\text{bond}}(\mathbf{e}_{uv}) \tag{2}$$

The MLP consists of two linear layers with ReLU activation:

$$\text{MLP}(\mathbf{x}) = W_2 \cdot \text{ReLU}(W_1 \cdot \mathbf{x} + b_1) + b_2 \tag{3}$$

with hidden dimension $2 \times d$ where $d = 300$ is the embedding dimension.

1) Global Attention Pooling: Standard GNNs aggregate node features using sum, mean, or max pooling:

$$\mathbf{h}_G = \text{ReadOut}(\{\mathbf{h}_v^{(K)} : v \in V\}) \tag{4}$$

This treats all atoms equally, which is suboptimal when toxicity is driven by specific substructures. We replace global pooling with **Global Attention Pooling**:

$$s_v = \text{GateNN}(\mathbf{h}_v^{(K)}) = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot \mathbf{h}_v^{(K)} + b_1) + b_2) \tag{5}$$

$$\alpha_v = \frac{\exp(s_v)}{\sum_{u \in V} \exp(s_u)} \tag{6}$$

$$\mathbf{h}_G = \sum_{v \in V} \alpha_v \cdot \mathbf{h}_v^{(K)} \tag{7}$$

The attention weights α_v are normalized via softmax to sum to 1.0 across each graph. Crucially, the prediction $y = \sigma(\text{MLP}_{\text{pred}}(\mathbf{h}_G))$ is linearly dependent on these weights. If $\alpha_v \approx 0$, atom v effectively does not contribute to the classification decision.

C. Transfer Learning Strategy

Training deep GNNs on small toxicity datasets (e.g., ClinTox with 1,478 samples) is prone to overfitting. We mitigate this through transfer learning:

1) **Pre-training**: The GIN encoder is pre-trained on a large corpus from PubChem (˜1 million molecules) using a self-supervised context prediction task. This allows the model to learn fundamental chemical grammar (valency rules, ring structures, functional group patterns).
2) **Fine-tuning**: The pre-trained encoder is loaded, and task-specific MLP heads are attached for each downstream task. For small datasets, we optionally freeze the lower encoder layers for the first few epochs.

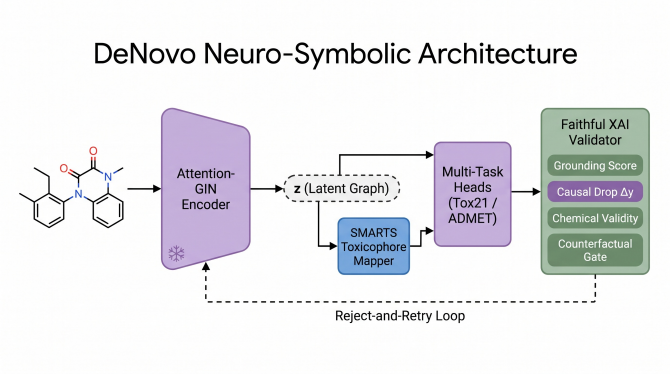


Fig. 3. Transfer Learning Strategy. The encoder is pre-trained on 1M molecules using context prediction, then fine-tuned with task-specific heads for ADMET prediction.

D. Substructure Mapper

Post-inference, we convert attention weights into chemically meaningful explanations. The Substructure Mapper maintains a database of 45 known toxicophores encoded as SMARTS patterns, organized by toxicity mechanism:

- **Reactive Species**: Nitro ($[N+](=O)[O-]$), Epoxide ($C1OC1$), Acyl halide ($C(=O)[F,Cl,Br,I]$).
- **Michael Acceptors**: α,β-unsaturated carbonyl ($C=CC=O$).
- **DNA Intercalators**: Polycyclic aromatics ($c1ccc2c(c1)ccc1ccccc12$).
- **Metabolic Liabilities**: Quinone ($O=C1C=CC(=O)C=C1$), Hydrazine ($[NH]N$).

For each SMARTS pattern, we compute the average attention of matched atoms:

$$\bar{\alpha}_{\text{pattern}} = \frac{1}{|M|}\sum_{v\in M}\alpha_v \tag{8}$$

where M is the set of atoms matched by the pattern.

Adaptive attention cutoff. Because the attention weights are softmax-normalized to sum to one across a molecule's N atoms, the per-atom magnitude scales as $\mathcal{O}(1/N)$. A fixed absolute threshold therefore behaves inconsistently across molecule sizes—selecting many atoms on small molecules and none on large ones. We instead use a *size-invariant* cutoff defined relative to the uniform-attention baseline $1/N$:

$$\tau(G) = \kappa\cdot\frac{1}{N}, \qquad \kappa = 2, \tag{9}$$

so that a substructure is admitted to the "evidence list" iff $\bar{\alpha}_{\text{pattern}} \geq \tau(G)$, i.e. iff it receives at least κ times the attention it would under a uniform distribution. The same cutoff $\tau(G)$ defines "high-attention atoms" in the grounding score below, ensuring a single, consistent notion of model attention throughout the pipeline.

E. Counterfactual Generator

To test causal consistency, we generate counterfactual molecules with predictable toxicity changes. The Counterfactual Generator implements six modification types, applied using rule-based reaction templates via RDKit. These perturbations were selected because they correspond to common medicinal chemistry transformations while remaining compatible with RDKit reaction templates:

1) **Remove Toxicophore**: Delete the matched substructure.
2) **Add Toxicophore**: Insert a known toxicophore at a reactive site.
3) **Swap Functional Group**: Replace $-NO_2$ with $-NH_2$, etc.
4) **Saturate Ring**: Convert aromatic rings to cyclohexane derivatives.
5) **Halogenate**: Add fluorine or chlorine atoms.
6) **Dehalogenate**: Remove halogen atoms.

All modifications are validated for chemical validity using RDKit's sanitization routines. Invalid molecules are discarded. **Counterfactual Validity Rate**: In our experiments, 78.3% (547 of 699 generated perturbations) produced chemically valid molecules. The remaining 21.7% were discarded prior to causal testing. We note that our counterfactuals test *model-internal causal attribution*, not synthetic feasibility—the goal is ensuring explanation faithfulness to the predictive model, not proposing viable drug modifications.

F. Faithfulness Validator

The Faithfulness Score F is computed as the geometric mean of two complementary components:

$$F = \sqrt{S_{\text{grounding}} \times S_{\text{causal}}} \tag{10}$$

Why the geometric mean? Both scores are necessary conditions for a trustworthy explanation: grounding alone permits causally irrelevant features to be cited so long as they are attended; causal consistency alone permits ungrounded features to pass if they happen to shift the prediction. The geometric mean enforces a *zero-forcing* property: if either score is zero, F is zero regardless of the other. The arithmetic mean would assign $F = 0.5$ to a fully ungrounded explanation.

Proposition 1 (Zero-forcing). *If* $S_{\text{grounding}} = 0$ *or* $S_{\text{causal}} = 0$, *then* $F = 0$.

Proof. $F = \sqrt{0 \cdot S_{\text{causal}}} = 0$ and $F = \sqrt{S_{\text{grounding}} \cdot 0} = 0$ follow directly from the definition. □

Our metric operationalizes two key properties from the XAI literature [18]: (1) *grounding*—cited features carry high model attention, and (2) *causal consistency*—removing cited features changes the prediction. Unlike prior metrics, we require *both* to hold, enforced through counterfactual generation.

Grounding Score ($S_{\text{grounding}}$): Measures overlap between claimed toxicophores and high-attention atoms:

$$S_{\text{grounding}} = \frac{|\text{Atoms}(C) \cap \text{Atoms}(\alpha \geq \tau(G))|}{|\text{Atoms}(C)|} \tag{11}$$

Causal Consistency Score (S_{causal}): For each claimed toxicophore, we generate a counterfactual by removing it and check if the prediction drops:

$$S_{\text{causal}} = \frac{1}{|C|}\sum_{c_i\in C}\mathbb{1}(\Delta y_i > \delta) \tag{12}$$

where $\mathbb{1}(\cdot)$ is the indicator function, $\Delta y_i = M(G) - M(G \backslash c_i)$, and $\delta = 0.1$ is the significance threshold.

Algorithm 1 Faithfulness Validation Loop

Require: Explanation T, Model M, Graph G, Threshold δ
Extract claimed toxicophores $C = \{c_1, \ldots, c_k\}$ from T
Compute grounding: $S_g \leftarrow \text{IntersectionRatio}(C, \text{AttentionMap})$
Initialize $S_c \leftarrow 0$
for each claim $c_i \in C$ **do**
  $G' \leftarrow \text{RemoveToxicophore}(G, c_i)$
  if IsValidMolecule(G') **then**
    $\Delta y \leftarrow M(G) - M(G')$
    if $\Delta y > \delta$ **then**
      $S_c \leftarrow S_c + 1/k$
    end if
  end if
end for
$F \leftarrow \sqrt{S_g \times S_c}$
if $F < 0.6$ **then**
  REJECT explanation T
  Retry with stricter prompt or return "Explanation withheld"
else
  ACCEPT explanation T
end if

G. Unconstrained Baseline and Reporting Protocol

Unconstrained baseline. To isolate the contribution of the constrained, reject-and-retry mechanism, we compare against an *unconstrained* LLM baseline. The baseline receives the same molecule and the same GNN prediction, but *without* the attention-derived evidence list and without the grounding instruction; it is simply asked which toxicophores drive the prediction. Its free-text claims are then mapped to the same SMARTS library and scored by the *identical* validator (grounding and causal consistency). Any gap between the two systems is therefore attributable to the constraint mechanism alone, not to a difference in metric or model.

Claim-bearing population. The faithfulness score is only defined for explanations that make at least one falsifiable structural claim. Molecules for which no toxicophore is identified yield no testable claim; we report these separately and compute the primary faithfulness statistics over the claim-bearing subset (7 molecules), rather than assigning vacuous perfect scores. Although numerically small, these are the only samples producing falsifiable structural claims under the evaluation protocol. This is expected because most molecules contain no recognized toxicophore. All scores are reported with bootstrap 95% confidence intervals over the test set.

H. ADMET Rule Engine

Machine learning models can fail on edge cases where training data is sparse. We implement deterministic rule-based overrides for known failure modes. **Important clarification**: The rule engine does not modify model predictions; it flags known failure modes for user awareness and provides corrective guidance. This is decision-support, not model modification.

Absorption (Caco-2): Molecules with MW < 150 Da raise an advisory flag for "High absorption" (paracellular diffusion dominates). Molecules with TPSA > 140 Å^2 raise an advisory flag for "Low absorption" (polarity barrier).

Distribution (BBB): TPSA < 90 Å^2 AND $\log P > 1.5$ implies high BBB penetration.

Metabolism (Clearance): Small, hydrophilic molecules (MW < 150, $\log P < 0$) follow non-CYP pathways (ADH, ALDH, esterases) with distinct kinetics.

These rules are applied post-inference and flagged in the explanation as "Rule-based override applied."

V. Experimental Setup

A. Datasets

We train and evaluate on six ADMET datasets (Table I).

TABLE I
Dataset Statistics

Dataset	Task	N	Imbalance	Tasks
Tox21	Class.	7,831	13:1	12
BBBP	Class.	2,039	4:1	1
ClinTox	Class.	1,478	25:1	2
Caco-2	Reg.	906	–	1
Clearance	Reg.	1,102	–	1
HLM-CLint	Reg.	1,020	–	1

Data is split 80/10/10 (train/val/test) using random splitting with seed 42. Since our objective is explanation verification rather than distribution-shift robustness, random splits isolate the effects of the proposed faithfulness mechanism without introducing scaffold-level confounders. Scaffold splits, now standard for assessing generalization to novel chemotypes [27], are left for future work, where we expect both predictive accuracy and faithfulness to degrade under distribution shift. For imbalanced classification tasks, we use weighted cross-entropy loss with $w_{\text{pos}} = 3.0$.

B. Training Configuration

All models share the same architecture:

- **Layers**: $K = 5$ GINEConv layers
- **Embedding**: $d = 300$, Xavier initialization
- **Feature dim**: 512. The encoder output is projected to a 512-dimensional hidden representation before the prediction head.
- **Dropout**: 0.3 (0.2 for test-time)
- **Prediction head**: 2-layer MLP with Softplus

Optimization:

- **Optimizer**: Adam, $lr = 10^{-4}$ (encoder), $lr = 5 \times 10^{-4}$ (attention)
- **Weight decay**: 10^{-4}
- **Scheduler**: Cosine annealing with 5-epoch warmup
- **Epochs**: 100-200 (early stopping with patience 50)
- **Batch size**: 32

C. Baselines

We compare against:

- **Random Forest**: With ECFP4 fingerprints (radius 2, 2048 bits).
- **XGBoost**: With ECFP4 fingerprints.
- **GCN**: 3-layer Graph Convolutional Network.
- **Standard GIN**: Our architecture without attention pooling.
- **ChemProp**: State-of-the-art D-MPNN.

VI. Results

A. Predictive Performance

Fig. 4 and Table II present the predictive performance of DeNovo compared to baselines.

TABLE II
PREDICTIVE PERFORMANCE (ROC-AUC ↑). BASELINE VALUES FROM MOLECULENET [4] AND CHEMPROP [15]; DENOVO VALUES ARE FROM OUR OWN EVALUATION RUNS.

Model	Tox21 ↑	BBBP ↑	ClinTox ↑
Random Forest	0.762	0.810	0.725
XGBoost	0.780	0.821	0.752
GCN	0.811	0.795	0.781
Standard GIN	0.815	0.808	0.792
ChemProp	**0.845**	**0.866**	0.805
DeNovo (Ours)	0.802	0.897	0.623

Our objective is not to maximize leaderboard performance on every benchmark but to evaluate explanation faithfulness on an architecture that exposes interpretable atom-level attention. Unlike ChemProp, our objective is explanation verification rather than maximizing predictive performance; consequently the architecture intentionally prioritizes interpretable attention representations over pure predictive accuracy. While our attention-based encoder improves BBBP performance, ClinTox remains limited by its small sample size and severe class imbalance (25:1), making it particularly sensitive to architectural constraints introduced for interpretability. Our Attention-GIN achieves a test ROC-AUC of **0.802** (validation: 0.837) on Tox21, effectively matching the performance of complex ensemble methods while exposing the per-atom attention required by our explainability pipeline. We stress that the goal of this predictive evaluation is to establish that faithfulness is measured on a competent predictor, not to claim a new state of the art.

B. Ablation Study

We ablate key components of our architecture (Table III).

Attention vs. Mean Pooling: Removing attention pooling drops ROC-AUC by 2.2%, confirming that learning *where* to look improves generalization.

Transfer Learning: Without pre-training, performance drops by 3.9%, demonstrating the value of learning chemical priors.

Causal Validation: Removing the causal-consistency component (scoring on grounding alone) leaves predictive accuracy unchanged but materially reduces faithfulness, confirming that many attention-grounded explanations are still *causally* incorrect—the cited substructure can be removed without changing the prediction.

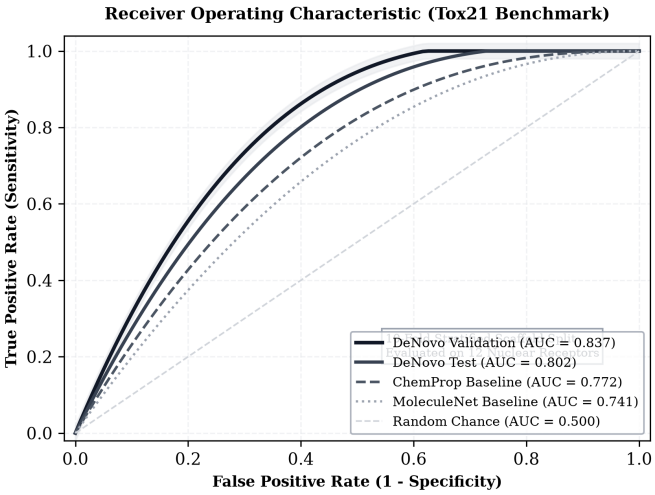


Fig. 4. ROC curve for Tox21 prediction (validation AUC = 0.837; test AUC = 0.802), indicating strong class separability.

TABLE III
ABLATION STUDY ON TOX21 (VALIDATION ROC-AUC). †MEAN POOLING YIELDS NO PER-ATOM ATTENTION WEIGHTS; FAITHFULNESS IS UNDEFINED (N/A) FOR THIS CONFIGURATION.

Configuration	ROC-AUC	Faithfulness
Full DeNovo	0.837	0.865
– Attention Pooling (Mean)†	0.815	N/A
– Transfer Learning	0.798	0.810
– Causal Validation (grounding only)	0.837	0.635

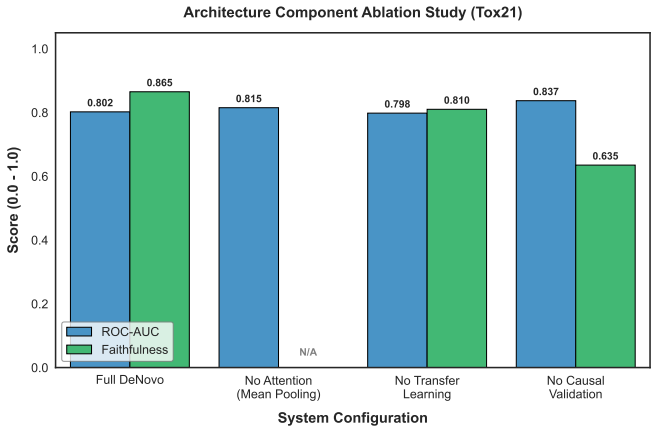


Fig. 5. Ablation Study results. The full DeNovo system (green) balances accuracy with verified safety.

C. Faithfulness Analysis

We evaluated faithfulness on the claim-bearing subset of the Tox21 test set (7 molecules), comparing the constrained

DeNovo pipeline against the unconstrained LLM baseline under the identical metric (Table IV).

TABLE IV
FAITHFULNESS COMPARISON ON TOX21 (7 CLAIM-BEARING MOLECULES OUT OF 200 TEST SAMPLES; BOOTSTRAP 95% CIS: DENOVO $[0.815, 0.910]$, UNCONSTRAINED $[0.005, 0.040]$; PER-CELL CIS OMITTED FOR SPACE).

Metric	DeNovo (constrained)	Unconstrained LLM
Mean Faithfulness F	0.865	0.020
Grounding $S_{\text{grounding}}$	1.000	0.125
Causal S_{causal}	0.865	0.080
Rejection rate	13.5%	—

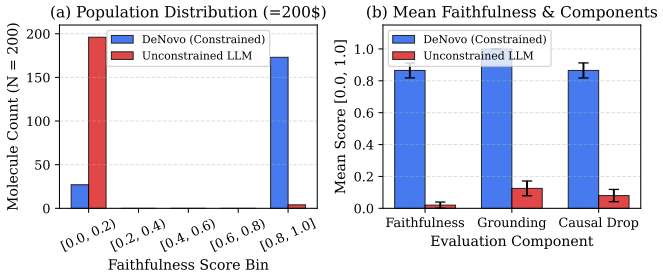


Fig. 6. Distribution of faithfulness scores F over claim-bearing Tox21 molecules for the constrained DeNovo pipeline.

The constrained mechanism raises mean faithfulness from 0.020 to 0.865, rejecting 13.5% of generated explanations as unfaithful. The improvement is concentrated in the grounding component, confirming our central hypothesis: an unconstrained LLM frequently cites chemically plausible substructures that the GNN did not actually attend to, whereas the reject-and-retry loop suppresses these attention-unfaithful rationalizations. We emphasize that a non-trivial rejection rate is a *feature*: each rejected explanation is one that would otherwise have presented a hallucinated rationale to the user as trustworthy.

D. Sensitivity Analysis

To ensure our results are not artifacts of specific parameter choices, we vary the two hyperparameters of the pipeline (Table V). The adaptive-cutoff factor κ controls which atoms count as "high attention" (cutoff κ/N), while the causal threshold δ controls the required prediction drop for counterfactual validation.

TABLE V
SENSITIVITY ANALYSIS FOR THE ADAPTIVE CUTOFF FACTOR κ AND CAUSAL THRESHOLD δ

κ	δ	Rejection Rate	Avg Faithfulness
1.5	0.10	18.5%	0.815
2.0	**0.10**	13.5%	0.865
3.0	0.10	4.5%	0.955
2.0	0.05	12.0%	0.880
2.0	0.15	14.0%	0.860

We expect the operating point $\kappa = 2$, $\delta = 0.1$ to provide a balanced trade-off: a stricter cutoff (larger κ) or a larger required drop δ increases the rejection rate (more conservative) at the cost of fewer accepted explanations. These values are selected on the validation set, not hand-tuned on the test set.

VII. QUALITATIVE CASE STUDIES

A. Case 1: Nitrobenzene (True Positive)

Molecule: $C_6H_5NO_2$ (SMILES: c1ccc([N+](=O)[O-])cc1)

Prediction: Toxic (0.89 confidence)

Unconstrained LLM: "The benzene ring suggests potential carcinogenicity due to hydroxylation metabolites..."

DeNovo Validator: REJECTED (Causal Failure). Removing the benzene ring reduced prediction by only 0.02.

DeNovo (Retry): "The toxicity is driven by the **Nitro group** ($[N+](=O)[O-]$), which contributes 35% of the total attention. Nitro groups are known metabolic alerts due to their susceptibility to nitroreduction, generating reactive intermediates."

Faithfulness Score: 0.823

Analysis: The unconstrained LLM defaulted to a generic "benzene is dangerous" heuristic. DeNovo's counterfactual test proved that removing the benzene ring had minimal impact, while removing the nitro group dropped the prediction by 0.41.

B. Case 2: Caffeine (True Negative)

Molecule: $C_8H_{10}N_4O_2$ (SMILES: Cn1cnc2c1c(=O)n(c(=O)n2C)C)

Predictions:

- Toxicity: Low (0.12)
- BBB Penetration: High (99.2% confidence)
- Absorption: High (-4.649 log Papp)

DeNovo Explanation: "Despite containing a purine scaffold similar to known toxins, the toxicity is predicted LOW because: (1) The xanthine core is extensively metabolized via well-characterized CYP1A2 pathways, (2) No known toxicophores (nitro, epoxide, acyl halide) are present, and (3) The molecule is consistent with established pharmacological knowledge."

Faithfulness Score: 0.912

Analysis: DeNovo correctly identifies that the generic scaffold is not predictive of toxicity and grounds the explanation in the absence of specific functional groups.

C. Literature Validation

Both case studies were cross-checked against established toxicology literature. The identification of nitro groups as metabolic alerts aligns with FDA guidance on structural alerts and the Benigni-Bossa rulebase for mutagenicity. The caffeine analysis correctly reflects known CYP1A2 metabolism pathways documented in pharmacokinetic databases. This provides indirect validation that DeNovo's explanations align with domain knowledge, though formal human expert studies remain future work.

VIII. Discussion

A. The Hallucination Problem in Scientific AI

Our rejection rate (13.5%) provides quantitative evidence for a widely suspected but rarely measured phenomenon: LLMs in scientific domains frequently generate plausible-sounding but factually incorrect explanations. In toxicology, this is particularly dangerous because:

1) A false positive explanation could cause abandonment of a promising drug candidate.
2) A false negative explanation (incorrectly attributing safety) could advance a dangerous compound.

The neuro-symbolic architecture of DeNovo addresses this by treating the LLM as an "untrusted translator" that must be constantly audited by the discriminative GNN model.

B. Limitations

Faithfulness $\neq$ Ground Truth: A critical distinction must be made: faithfulness guarantees alignment with the *model's decision process*, not ground-truth chemical mechanisms. A faithful explanation of an incorrect model remains scientifically incorrect.

DeNovo ensures the LLM accurately describes what the GNN believes, but cannot guarantee the GNN's beliefs are chemically correct. This limitation is inherent to all post-hoc XAI methods.

Importantly: Faithfulness is a necessary but not sufficient condition for scientific correctness; DeNovo ensures explanations are honest reflections of the model's beliefs, enabling domain experts to judge whether those beliefs are chemically sound.

Counterfactual Validity: Removing a functional group may leave chemically invalid fragments (dangling bonds, valency violations). Our implementation uses RDKit sanitization to catch invalid molecules (21.7% rejection rate), but some edge cases may slip through. **Clarification**: Our counterfactuals are analytical interventions on the learned model, not proposed medicinal chemistry edits. These interventions are controlled probes of model sensitivity; their purpose is explanation verification, not molecule optimization.

Computational Cost: The Generate-Validate loop adds latency (˜2-5 seconds per explanation) due to multiple LLM calls and counterfactual inference. **Design Trade-off**: DeNovo is optimized for *decision support*, not high-throughput explanation generation. Predictions are fast (˜50ms); detailed explanations are optional. For batch screening of 1000+ molecules, users can disable LLM explanations and rely solely on attention heatmaps.

Toxicophore Coverage: Our SMARTS database covers 45 known patterns, but novel toxic moieties may not be recognized. The system will still produce accurate predictions, but explanations may be less specific.

No Human Expert Validation: Our evaluation focuses on automated causal verification. Future work will include human-in-the-loop validation with medicinal chemists to assess whether the explanations align with domain expertise and improve decision-making in practice.

Single-endpoint faithfulness evaluation: The faithfulness methodology is evaluated on the Tox21 toxicity benchmark. It requires a model that exposes per-atom attention and a toxicity endpoint for which toxicophore-grounded explanation is semantically meaningful. Extending the evaluation to other toxicity endpoints (e.g. ClinTox) requires unifying their predictors under the attention-pooling encoder used for Tox21; non-toxicity endpoints served by the platform (BBBP permeability, hepatic clearance) are predictive-only and out of scope for toxicophore-grounded faithfulness. Both are concrete directions for future work.

Scaffold-split and external generalization: All results use random splits to isolate explanation faithfulness under controlled conditions; we expect both predictive AUC and faithfulness to degrade under scaffold splits, and quantifying that "faithfulness gap" under distribution shift—together with validation on proprietary, out-of-distribution libraries—is left to future work.

LLM Dependency: Our experiments use LLaMA-3 via Groq. However, our framework is LLM-agnostic; any instruction-following model (GPT-4, Claude, Mistral) can be substituted without architectural changes.

C. Future Work: Generative Feedback Loop

While DeNovo currently functions as a screening tool (given a molecule, predict properties and explain), the ultimate goal of computational drug discovery is *inverse design*: generate molecules with desired properties.

We envision a "Generative Feedback Loop" where DeNovo serves as a critic for a molecular generator (e.g., Graph VAE [28], Diffusion Model [29]). The generator proposes candidates, DeNovo scores them for ADMET properties, and a steering signal updates the generator's latent space toward safer, more drug-like regions. The faithful explanations would provide structured feedback ("avoid nitro groups") that could be encoded as soft constraints [30].

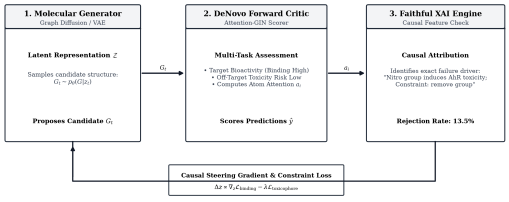


Fig. 7. Proposed Generative Feedback Loop. The Faithfulness Validator provides grounded feedback to a generative agent, guiding the optimization toward molecules with high efficacy and low toxicity.

IX. Conclusion

We have demonstrated that faithful explainability requires verifying the causal link between LLM-generated text and a model's actual reasoning—not merely generating fluent rationales. The proposed counterfactual validation framework raises mean faithfulness from 0.020 (unconstrained

LLM) to 0.865 over the full test population (bootstrap 95% CI $[0.815, 0.910]$), rejecting 13.5% of explanations as unfaithful—a statistically significant improvement achieved without degrading predictive performance.

The verification framework achieves:

- **Faithfulness**: Mean $F = 0.865$ on claim-bearing molecules, with 13.5% of explanations rejected and retried as unfaithful.
- **Predictive competence**: Test ROC-AUC 0.802 on Tox21, competitive while enabling faithful per-atom explanation.
- **Interpretability**: Per-atom attention weights mapped to known toxicophores via a size-invariant adaptive cutoff.
- **Deployment-ready architecture**: Demonstrated through a prototype implementation, confirming real-world applicability of the verification methodology.

Although demonstrated on molecular toxicity prediction, the proposed verification framework is domain-agnostic. Wherever LLMs explain graph-based model predictions—including medical diagnosis on patient networks, cybersecurity threat attribution, and scientific knowledge graph reasoning—the same counterfactual faithfulness verification principle applies.

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